# R, Databases & SQL

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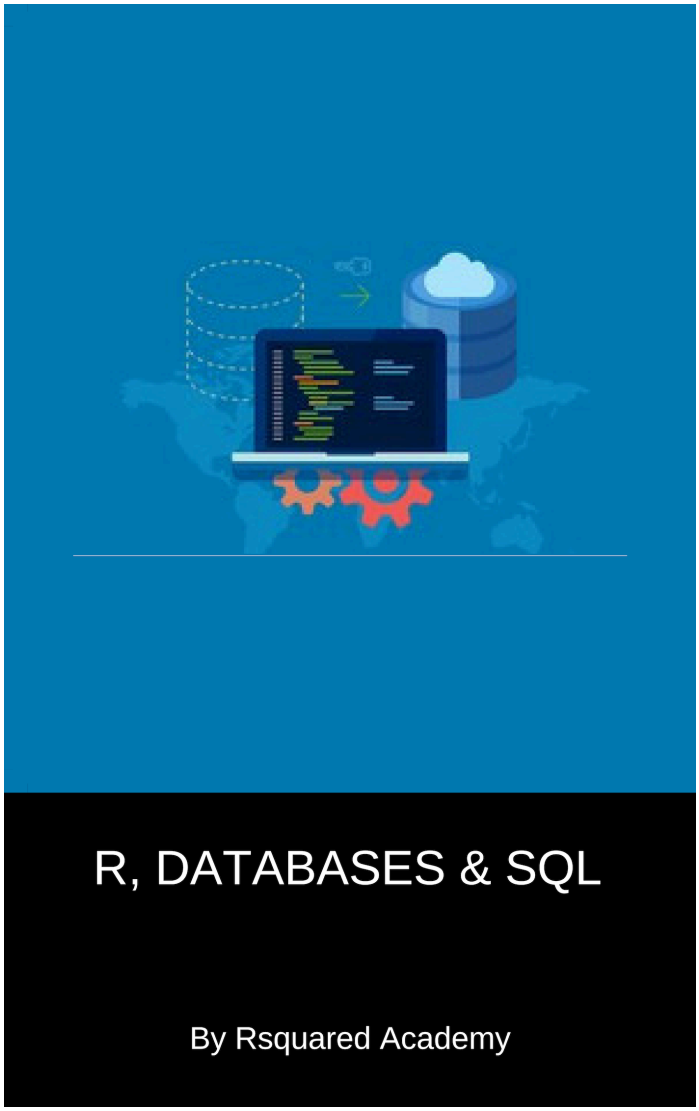
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# Preface

*Last verified: 2026-09-26 · R 4.5.2 · DBI · dbplyr 2.6.0 · RSQLite 2.4.6 · duckdb.*

New here? Grab the one-page DBI cheatsheet, then work Chapters 1–5 in order — runnable code for each chapter lives in `code/`.




## Structure of the book

Chapter Chapter 1 introduces the **DBI** package. Chapter Chapter 2 explores data wrangling within database using the **dplyr** package. Chapters Chapter 3 and Chapter 4 introduce basic and advanced SQL. Chapter Chapter 5 combines tables with JOINs in SQL and dbplyr.

## Software information

The R session information when compiling this book is shown below:

```
sessionInfo()
```

```
R version 4.4.2 (2024-10-31)
Platform: x86_64-pc-linux-gnu
Running under: Ubuntu 24.04.5 LTS
```

```
Matrix products: default
BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;
LAPACK version 3.12.0

locale:
 [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
 [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
 [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
[10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C

time zone: UTC
tzcode source: system (glibc)

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

loaded via a namespace (and not attached):
 [1] compiler_4.4.2  fastmap_1.2.0   cli_3.6.6       tools_4.4.2
 [5] htmltools_0.5.9 rmarkdown_2.32  knitr_1.52      jsonlite_2.0.0
 [9] xfun_0.61       digest_0.6.39   rlang_1.3.0     evaluate_1.0.5
```

We do not add prompts (> and +) to R source code in this book, and we comment out the text output with two hashes ## by default, as you can see from the R session information above. This is for your convenience when you want to copy and run the code (the text output will be ignored since it is commented out). Package names are in bold text (e.g., **rmarkdown**), and function names are followed by parentheses (e.g., `bookdown::render_book()`). The double-colon operator `::` means accessing an object from a package.

## About the Author

Aravind Hebbali is the founder of Rsquared Academy. He earned his Masters in Economics from Madras School of Economics. As an active R user, he has authored several R packages such as

- olsrr
- rfm
- descriptr
- blorr
- xplorerr

In 2015, he founded Rsquared Academy, a free and open source education initiative with focus on data science and analytics. Apart from self paced online courses, Rsquared Academy offers customized learning modules for corporates and universities.

You can find him on GitHub.

# 1 DBI

## 1.1 Introduction

In this chapter, we will learn to:

- connect to a SQLite database from R
- display database information
- list tables in the database
- query data
  - read entire table
  - read few rows
  - read data in batches
- create table in database
- overwrite table in database
- append data to table in database
- remove table from database
- generate SQL query
- close database connection

We will use the following R packages:

- DBI
- RSQLite

All the data sets used in this chapter can be found here and code can be downloaded from here or run from `code/ch1-dbi.R` in this repo.

## 1.2 Connection

The first step is to connect to a database. We will connect to an in memory SQLite databse using `dbConnect()`.

```
con <- dbConnect(RSQLite::SQLite(), ":memory:")
```

> **NOTE:** We use SQLite here because it needs zero setup — no server to install. The exact same DBI commands work with DuckDB, a high-speed analytical engine, for larger analytical queries (see R4DS Ch. 21 at scale):

```
# install.packages("duckdb")  # once; high-speed analytical engine
con_duck <- dbConnect(duckdb::duckdb())
dbWriteTable(con_duck, "ecom", ecom)
dbGetQuery(con_duck, "SELECT referrer, AVG(duration) FROM ecom GROUP BY referrer")
```

### 1.2.1 Connection Summary

We can get the more information about the connection using `summary()`.

```
summary(con)
```

```
          Length            Class             Mode
               1 SQLiteConnection               S4
```

### 1.2.2 List Tables

Now that we are connected to a database, let us list all the tables present in it using `dbListTables()`.

```
dbListTables(con)
```

```
[1] "ecom"         "sqlite_stat1" "sqlite_stat4"
```

### 1.2.3 List Fields

Time to explore the `ecom` table in the database. Use `dbListFields()` to list all the fields in the table.

```
dbListFields(con, "ecom")
```

```
[1] "referrer" "device"   "bouncers" "n_visit"  "n_pages"  "duration"
```

## 1.3 Querying Data

The main objectives of connecting to a database are to:

- query data from the tables already present
- create new tables
- overwrite existing tables
- delete existing tables

Let us begin with querying data. We can do this in the following ways:

- read an entire table at once
- read few rows from a table
- read data in batches

### 1.3.1 Entire Table

We can read an entire table from a database using `dbReadTable()`.

```
dbReadTable(con, 'ecom')
```

```
   referrer device bouncers n_visit n_pages duration
1    google laptop        1      10       1      693
2     yahoo tablet        1       9       1      459
3    direct laptop        1       0       1      996
4      bing tablet        0       3      18      468
5     yahoo mobile        1       9       1      955
6     yahoo laptop        0       5       5      135
7     yahoo mobile        1      10       1       75
8    direct mobile        1      10       1      908
9      bing mobile        0       3      19      209
10   google mobile        1       6       1      208
11   direct laptop        1       9       1      738
12   direct tablet        0       6      12      132
```

```
13   direct mobile        0       9      14      406
14    yahoo tablet        0       5       8       80
15    yahoo mobile        0       7       1       19
16     bing laptop        1       1       1      995
17     bing tablet        0       5      16      368
18   google tablet        1       7       1      406
19   social tablet        0       7      10      290
20   social tablet        0       2       1       28
```

In some cases, we may not need the entire table but only a specific number of rows. Use `dbGetQuery()` and supply a SQL statement specifying the number of rows of data to be read from the table. In the below example, we read ten rows of data from the `ecom` table.

### 1.3.2 Few Rows

```
dbGetQuery(con, "select * from ecom limit 10")
```

```
   referrer device bouncers n_visit n_pages duration
1    google laptop        1      10       1      693
2     yahoo tablet        1       9       1      459
3    direct laptop        1       0       1      996
4      bing tablet        0       3      18      468
5     yahoo mobile        1       9       1      955
6     yahoo laptop        0       5       5      135
7     yahoo mobile        1      10       1       75
8    direct mobile        1      10       1      908
9      bing mobile        0       3      19      209
10   google mobile        1       6       1      208
```

In case of very large table, we can read data in batches using `dbSendQuery()` and `dbFetch()`. We can mention the number of rows of data to be read while fetching the data using the query generated by `dbGetQuery()`.

### 1.3.3 Read Data in Batches

```
query <- dbSendQuery(con, 'select * from ecom')
result <- dbFetch(query, n = 15)
result
```

```
   referrer device bouncers n_visit n_pages duration
1    google laptop        1      10       1      693
2     yahoo tablet        1       9       1      459
3    direct laptop        1       0       1      996
4      bing tablet        0       3      18      468
5     yahoo mobile        1       9       1      955
6     yahoo laptop        0       5       5      135
7     yahoo mobile        1      10       1       75
8    direct mobile        1      10       1      908
9      bing mobile        0       3      19      209
```

```
10   google mobile        1       6       1      208
11   direct laptop        1       9       1      738
12   direct tablet        0       6      12      132
13   direct mobile        0       9      14      406
14    yahoo tablet        0       5       8       80
15    yahoo mobile        0       7       1       19
```

## 1.4 Query

### 1.4.1 Query Status

To know the status of a query, use `dbHasCompleted()`. It is very useful in cases of queries that might take a long time to complete.

```
dbHasCompleted(query)
```

```
[1] FALSE
```

### 1.4.2 Query Info

`dbGetInfo()` will return the following:

- the sql staement
- number of rows fetched
- number of rows modified/affected
- status of the query

```
dbGetInfo(query)
```

```
$statement
[1] "select * from ecom"

$row.count
[1] 15

$rows.affected
[1] 0

$has.completed
[1] FALSE
```

### 1.4.3 Latest Query

To get the latest query, use `dbGetStatement()`.

```
dbGetStatement(query)
```

```
[1] "select * from ecom"
```

### 1.4.4 Rows Fetched

To check the number of rows of data returned by a query, use `dbGetRowCount()`.

```
dbGetRowCount(query)
```

```
[1] 15
```

### 1.4.5 Rows Affected

To know the number of rows modified or affected in the table, use `dbGetRowsAffected()`.

```
dbGetRowsAffected(query)
```

```
[1] 0
```

### 1.4.6 Column Info

To know the name of the columns and their data types, use `dbColumnInfo()`.

```
dbColumnInfo(query)
```

```
      name      type
1 referrer character
2   device character
3 bouncers   integer
4  n_visit    double
5  n_pages    double
6 duration    double
```

Always clear the result set once you are done fetching and inspecting it.

```
dbClearResult(query)
```

## 1.5 Create Table

So far we have explored querying data from an existing table. Now, let us turn our attention to creating new tables in the database.

#### 1.5.0.1 Introduction

To create a new table, use `dbWriteTable()`. It takes the following 3 arguments:

- connection name
- name of the new table
- data for the new table

```
x <- 1:10
y <- letters[1:10]
trial <- tibble::tibble(x, y)
dbWriteTable(con, "trial", trial)
```

Let us check if the new table has been created.

```
dbListTables(con)
```

```
[1] "ecom"          "sqlite_stat1" "sqlite_stat4" "trial"
```

```
dbExistsTable(con, "trial")
```

```
[1] TRUE
```

Let us query data from the new table.

```
dbGetQuery(con, "select * from trial limit 5")
```

```
  x y
1 1 a
2 2 b
3 3 c
4 4 d
5 5 e
```

#### 1.5.0.2 Overwrite Table

In some cases, you may want to overwrite the data in an existing table. Use the `overwrite` argument in `dbWriteTable()` and set it to `TRUE`.

```
x <- sample(100, 10)
y <- letters[11:20]
trial2 <- tibble::tibble(x, y)
dbWriteTable(con, "trial", trial2, overwrite = TRUE)
```

Let us see if the **trial** table has been overwritten.

```
dbGetQuery(con, "select * from trial limit 5")
```

```
   x y
1 25 k
2 17 l
3 83 m
4 27 n
5 46 o
```

## 1.6 Append Data

You can append data to an existing table by setting the `append` argument in `dbWriteTable()` to `TRUE`.

```
x <- sample(100, 10)
y <- letters[5:14]
trial3 <- tibble::tibble(x, y)
dbWriteTable(con, "trial", trial3, append = TRUE)
```

Let us quickly check if the new data has been appended to the **trial** table.

```
dbReadTable(con, "trial")
```

```
    x y
1  25 k
2  17 l
3  83 m
4  27 n
5  46 o
6  34 p
```

```
7  65 q
8  59 r
9  90 s
10 52 t
11  4 e
12 84 f
13 32 g
14 53 h
15 65 i
16 98 j
17 77 k
18 45 l
19 34 m
20  2 n
```

We can also use `sqlAppendTable()` to append data to an existing table.

```
sqlAppendTable(con, "ecom", head(ecom))
```

```
<SQL> INSERT INTO `ecom`
  (`referrer`, `device`, `bouncers`, `n_visit`, `n_pages`, `duration`)
VALUES
  ('google', 'laptop', TRUE, 10, 1, 693),
  ('yahoo', 'tablet', TRUE, 9, 1, 459),
  ('direct', 'laptop', TRUE, 0, 1, 996),
  ('bing', 'tablet', FALSE, 3, 18, 468),
  ('yahoo', 'mobile', TRUE, 9, 1, 955),
  ('yahoo', 'laptop', FALSE, 5, 5, 135)
```

## 1.7 Insert Rows

#### 1.7.0.1 Introduction

We can insert new rows into existing tables using:

- `dbExecute()`
- `dbSendStatement()`

Both the function take 2 arguments:

- connection name
- sql statement

```
# use dbExecute
dbExecute(con,
  "INSERT into trial (x, y) VALUES (32, 'c'), (45, 'k'), (61, 'h')"
)
## [1] 3
```

```
# use dbSendStatement
res <- dbSendStatement(con,
  "INSERT into trial (x, y) VALUES (25, 'm'), (54, 'l'), (16, 'y')"
)
dbClearResult(res)
```

### 1.8 Remove Table

If you want to delete/remove a table from the database, use `dbRemoveTable()`.

```
dbRemoveTable(con, "trial")
```

### 1.9 SQLite Data Type

If you want to know the data type, use `dbDataType()`.

```
dbDataType(RSQLite::SQLite(), "a")
```

```
[1] "TEXT"
```

```
dbDataType(RSQLite::SQLite(), 1:5)
```

```
[1] "INTEGER"
```

```
dbDataType(RSQLite::SQLite(), 1.5)
```

```
[1] "REAL"
```

### 1.10 Close Connection

It is a good practice to close connection to a database when you no longer need to read/write data from/to it. Use `dbDisconnect()` to close the database connection.

```
dbDisconnect(con)
```

## 2 dbplyr

### 2.1 Introduction

In this chapter, we will learn to query data from a database using dplyr.

We will use the following R packages:

- DBI
- RSQLite
- dbplyr
- dplyr

All the data sets used in this chapter can be found here and code can be downloaded from here or run from `code/ch2-dbplyr.R` in this repo.

### 2.2 Connect to Database

Let us connect to an in memory SQLite database using `dbConnect()`.

```
con <- dbConnect(RSQLite::SQLite(), ":memory:")
```

We will copy the `ecom` data to the database so that we can use it for running dplyr statements. We read the same `ecom` data set used across chapters (1000 rows, 8 columns including `country` and `purchase`) from the local `data/web.csv` snapshot, falling back to the online copy if needed.

```
web_path <- if (file.exists("data/web.csv")) "data/web.csv" else "https://raw.git
hubusercontent.com/rsquaredacademy/datasets/master/web.csv"
ecom <- readr::read_csv(web_path, show_col_types = FALSE)
ecom <- dplyr::select(ecom, referrer, device, bouncers, n_visit, n_pages, duration,
country, purchase)
dplyr::copy_to(con, ecom)
```

## 2.3 Reference Data

In order to use dplyr functions, we need to reference the table in the database using `tbl()`.

```
ecom2 <- dplyr::tbl(con, "ecom")
ecom2
```

```
# A query:  ?? x 8
# Database: sqlite 3.53.3 [:memory:]
   referrer device bouncers n_visit n_pages duration country        purchase
   <chr>    <chr>     <int>   <dbl>   <dbl>    <dbl> <chr>             <int>
 1 google   laptop        1      10       1      693 Czech Republic        0
 2 yahoo    tablet        1       9       1      459 Yemen                 0
 3 direct   laptop        1       0       1      996 Brazil                0
 4 bing     tablet        0       3      18      468 China                 1
 5 yahoo    mobile        1       9       1      955 Poland                0
 6 yahoo    laptop        0       5       5      135 South Africa          0
 7 yahoo    mobile        1      10       1       75 Bangladesh            0
 8 direct   mobile        1      10       1      908 Indonesia             0
 9 bing     mobile        0       3      19      209 Netherlands           0
10 google   mobile        1       6       1      208 Czech Republic        0
# i more rows
```

## 2.4 Query Data

We will look at some simple examples. Let us start by selecting `referrer`, `device` and `duration` columns from `ecom2`.

```
select(ecom2, referrer, device, duration)
```

```
# A query:  ?? x 3
# Database: sqlite 3.53.3 [:memory:]
   referrer device duration
   <chr>    <chr>     <dbl>
 1 google   laptop      693
 2 yahoo    tablet      459
 3 direct   laptop      996
 4 bing     tablet      468
```

```
 5 yahoo    mobile      955
 6 yahoo    laptop      135
 7 yahoo    mobile       75
 8 direct   mobile      908
 9 bing     mobile      209
10 google   mobile      208
# i more rows
```

We can filter data as well. Filter all the rows from `ecom2` where `duration` is greater than 300.

```
filter(ecom2, duration > 300)
```

```
# A query:  ?? x 8
# Database: sqlite 3.53.3 [:memory:]
   referrer device bouncers n_visit n_pages duration country        purchase
   <chr>    <chr>     <int>   <dbl>   <dbl>    <dbl> <chr>             <int>
 1 google   laptop        1      10       1      693 Czech Republic        0
 2 yahoo    tablet        1       9       1      459 Yemen                 0
 3 direct   laptop        1       0       1      996 Brazil                0
 4 bing     tablet        0       3      18      468 China                 1
 5 yahoo    mobile        1       9       1      955 Poland                0
 6 direct   mobile        1      10       1      908 Indonesia             0
 7 direct   laptop        1       9       1      738 Jamaica               0
 8 direct   mobile        0       9      14      406 Ireland               1
 9 bing     laptop        1       1       1      995 United States         0
10 bing     tablet        0       5      16      368 Peru                  1
# i more rows
```

Time to do some grouping and summarizing. Let us compute the average time spent on the site for different types of devices.

```
ecom2 %>%
  group_by(device) %>%
  summarise(avg_duration = mean(duration))
```

```
# A query:  ?? x 2
# Database: sqlite 3.53.3 [:memory:]
  device avg_duration
  <chr>         <dbl>
1 laptop         376.
2 mobile         337.
3 tablet         354.
```

## 2.5 Show Query

If you want to view the SQL query generated in the above step, use `show_query()` or `explain()`.

```
avg_time <-
  ecom2 %>%
```

```
  group_by(device) %>%
  summarise(avg_duration = mean(duration))

dplyr::show_query(avg_time)
## <SQL>
## SELECT `device`, AVG(`duration`) AS `avg_duration`
## FROM `ecom`
## GROUP BY `device`

dplyr::explain(avg_time)
## <SQL>
## SELECT `device`, AVG(`duration`) AS `avg_duration`
## FROM `ecom`
## GROUP BY `device`
##
## <PLAN>
##   id parent notused                    detail
## 1  6      0     115                 SCAN ecom
## 2  8      0       0 USE TEMP B-TREE FOR GROUP BY
```

### 2.6 Collect Data

Now, some interesting facts. When working with databases, **dplyr** never pulls data into R unless you explicitly ask for it. In the previous example, dplyr will not do anything until you ask for the `avg_time` data. It generates the SQL and only pulls down a few rows when you try to print `avg_time`. So how do we pull all the data and store it for further analysis? `collect()` will pull all the data and store it in a tibble and you can use it for any further analysis.

```
dplyr::collect(avg_time)
```

```
# A tibble: 3 x 2
  device avg_duration
  <chr>         <dbl>
1 laptop         376.
2 mobile         337.
3 tablet         354.
```

It is good practice to close the connection when you are done:

```
dbDisconnect(con)
```

## 3 SQL Basics

### 3.1 Introduction

In this chapter, we will learn to:

- select
  - single column

  - multiple columns
  - distinct values in a column
- limit the number of records returned
- handle NULL values
- and filter columns using the following operators
  - WHERE
  - AND, or & NOT
  - BETWEEN
  - IN
  - LIKE

We will use the following R packages:

- DBI
- RSQLite
- dbplyr

All the data sets used in this chapter can be found here and code can be downloaded from here or run from `code/ch3-sqlbasics.R` in this repo.

## 3.2 Set Up

In order to be able to use the `ecom` data set for our case study, we will read it using the **readr** package and then copy it to an in memory database using the `copy_to()` function from **dplyr**.

```
# Canonical ecom: 1000 rows x 8 cols (referrer, device, bouncers, n_visit,
# n_pages, duration, country, purchase), from the local data/web.csv snapshot.
web_path <- if (file.exists("data/web.csv")) "data/web.csv" else "https://raw.git
hubusercontent.com/rsquaredacademy/datasets/master/web.csv"
ecom <- readr::read_csv(web_path, show_col_types = FALSE)
ecom <- dplyr::select(ecom, referrer, device, bouncers, n_visit, n_pages, duration,
country, purchase)
con <- DBI::dbConnect(RSQLite::SQLite(), ":memory:")
dplyr::copy_to(con, ecom)
```

## 3.3 Select Columns

The SQL SELECT statement is used to fetch the data from a database table.

### 3.3.1 Syntax

Below is the basic syntax of the SELECT statement.

```
SELECT column1, column2, columnN FROM table_name;
```

Here, column1, column2... are the fields of a table whose values you want to fetch. If you want to fetch all the fields, use the following syntax.

```
SELECT * FROM table_name;
```

### 3.3.2 Select Single Column

Let us begin by selecting the `device` field from the `ecom` table.

```
SELECT device FROM ecom
```

| device |
|---|
| laptop |
| tablet |
| laptop |
| tablet |
| mobile |
| laptop |
| mobile |
| mobile |
| mobile |
| mobile |

### 3.3.3 Select Multiple Columns

Select the following fields from the `ecom` table:

- `referrer`
- `device`
- `purchase`

```
SELECT referrer, device, purchase  FROM ecom
```

| referrer | device | purchase |
|---|---|---|
| google | laptop | 0 |
| yahoo | tablet | 0 |
| direct | laptop | 0 |
| bing | tablet | 1 |
| yahoo | mobile | 0 |
| yahoo | laptop | 0 |
| yahoo | mobile | 0 |
| direct | mobile | 0 |
| bing | mobile | 0 |
| google | mobile | 0 |

### 3.3.4 Select All Columns

Select all the fields from the `ecom` table.

```
SELECT * FROM ecom
```

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| google | laptop | 1 | 10 | 1 | 693 | Czech Republic | 0 |
| yahoo | tablet | 1 | 9 | 1 | 459 | Yemen | 0 |
| direct | laptop | 1 | 0 | 1 | 996 | Brazil | 0 |
| bing | tablet | 0 | 3 | 18 | 468 | China | 1 |
| yahoo | mobile | 1 | 9 | 1 | 955 | Poland | 0 |
| yahoo | laptop | 0 | 5 | 5 | 135 | South Africa | 0 |
| yahoo | mobile | 1 | 10 | 1 | 75 | Bangladesh | 0 |
| direct | mobile | 1 | 10 | 1 | 908 | Indonesia | 0 |
| bing | mobile | 0 | 3 | 19 | 209 | Netherlands | 0 |
| google | mobile | 1 | 6 | 1 | 208 | Czech Republic | 0 |

## 3.4 Limit

If you have a large table with thousands of rows, returning all the records will take time. Use LIMIT to specify the number of records to return.

```
SELECT * FROM ecom limit 10
```

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| google | laptop | 1 | 10 | 1 | 693 | Czech Republic | 0 |
| yahoo | tablet | 1 | 9 | 1 | 459 | Yemen | 0 |
| direct | laptop | 1 | 0 | 1 | 996 | Brazil | 0 |
| bing | tablet | 0 | 3 | 18 | 468 | China | 1 |
| yahoo | mobile | 1 | 9 | 1 | 955 | Poland | 0 |
| yahoo | laptop | 0 | 5 | 5 | 135 | South Africa | 0 |

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| yahoo | mobile | 1 | 10 | 1 | 75 | Bangladesh | 0 |
| direct | mobile | 1 | 10 | 1 | 908 | Indonesia | 0 |
| bing | mobile | 0 | 3 | 19 | 209 | Netherlands | 0 |
| google | mobile | 1 | 6 | 1 | 208 | Czech Republic | 0 |

## 3.5 Distinct

A column in a table may often contain many duplicate values; and we might be interested only in the distinct/unique values. In such cases, we can use the SELECT DISTINCT statement to return only distinct values.

```
SELECT distinct referrer FROM ecom
```

| referrer |
|---|
| google |
| yahoo |
| direct |
| bing |
| social |

## 3.6 Filter

Now that we know how to select columns, let us focus on filtering data. In SQL, the WHERE keyword is used to extract only those records that fulfill a specified condition. Data filter based on both text and numeric values in a table. Below are a few comparison operators we can use:

- `=` equal
- `<>` not equal
- `<` less than
- `>` greater than
- `<=` less than or equal to
- `>=` greater than or equal to

The following SQL statement filters all rows from the `ecom` table where the `duration` field is greater than 300.

```
SELECT *
FROM ecom
WHERE duration > 300
```

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| google | laptop | 1 | 10 | 1 | 693 | Czech Republic | 0 |
| yahoo | tablet | 1 | 9 | 1 | 459 | Yemen | 0 |
| direct | laptop | 1 | 0 | 1 | 996 | Brazil | 0 |
| bing | tablet | 0 | 3 | 18 | 468 | China | 1 |
| yahoo | mobile | 1 | 9 | 1 | 955 | Poland | 0 |
| direct | mobile | 1 | 10 | 1 | 908 | Indonesia | 0 |
| direct | laptop | 1 | 9 | 1 | 738 | Jamaica | 0 |
| direct | mobile | 0 | 9 | 14 | 406 | Ireland | 1 |
| bing | laptop | 1 | 1 | 1 | 995 | United States | 0 |
| bing | tablet | 0 | 5 | 16 | 368 | Peru | 1 |

Let us filter data based on a text value. In the following example, we filter all rows from the `ecom` table where the `device` used is `mobile`.

```
SELECT *
FROM ecom
WHERE device = 'mobile'
```

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| yahoo | mobile | 1 | 9 | 1 | 955 | Poland | 0 |
| yahoo | mobile | 1 | 10 | 1 | 75 | Bangladesh | 0 |
| direct | mobile | 1 | 10 | 1 | 908 | Indonesia | 0 |
| bing | mobile | 0 | 3 | 19 | 209 | Netherlands | 0 |

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| google | mobile | 1 | 6 | 1 | 208 | Czech Republic | 0 |
| direct | mobile | 0 | 9 | 14 | 406 | Ireland | 1 |
| yahoo | mobile | 0 | 7 | 1 | 19 | France | 0 |
| google | mobile | 1 | 5 | 1 | 147 | Brazil | 0 |
| bing | mobile | 0 | 0 | 7 | 196 | Russia | 0 |
| google | mobile | 1 | 10 | 1 | 338 | Russia | 0 |

## 3.7 And, Or & Not

The WHERE clause can be combined with other operators such as

- AND - displays a record if all the conditions separated by AND is TRUE
- OR - displays a record if any of the conditions separated by OR is TRUE
- NOT - displays a record if the condition(s) is NOT TRUE

to filter data based on more than one condition or to create more complex conditions.

In the following example, we filter all the rows from the `ecom` table where `n_visit` (visit count) is greater than 3 and `duration` (time spent on the site) is greater than 100. We use AND to create multiple conditions.

```
SELECT *
FROM ecom
WHERE n_visit > 3 AND duration > 100
```

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| google | laptop | 1 | 10 | 1 | 693 | Czech Republic | 0 |
| yahoo | tablet | 1 | 9 | 1 | 459 | Yemen | 0 |
| yahoo | mobile | 1 | 9 | 1 | 955 | Poland | 0 |
| yahoo | laptop | 0 | 5 | 5 | 135 | South Africa | 0 |

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| direct | mobile | 1 | 10 | 1 | 908 | Indonesia | 0 |
| google | mobile | 1 | 6 | 1 | 208 | Czech Republic | 0 |
| direct | laptop | 1 | 9 | 1 | 738 | Jamaica | 0 |
| direct | tablet | 0 | 6 | 12 | 132 | Estonia | 0 |
| direct | mobile | 0 | 9 | 14 | 406 | Ireland | 1 |
| bing | tablet | 0 | 5 | 16 | 368 | Peru | 1 |

In the next example, we will use both AND & OR. Our goal is to filter all rows from the `ecom` table that follow the below conditions:

- `n_visit` (visit count) is either equal to 3 or 5
- `device` used to visit the website is either `mobile` or `tablet`

```
SELECT *
FROM ecom WHERE (n_visit = 5 OR n_visit = 3)
AND (device = 'mobile' OR device = 'tablet')
```

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| bing | tablet | 0 | 3 | 18 | 468 | China | 1 |
| bing | mobile | 0 | 3 | 19 | 209 | Netherlands | 0 |
| yahoo | tablet | 0 | 5 | 8 | 80 | Philippines | 0 |
| bing | tablet | 0 | 5 | 16 | 368 | Peru | 1 |
| google | mobile | 1 | 5 | 1 | 147 | Brazil | 0 |
| yahoo | mobile | 1 | 3 | 1 | 700 | Canada | 0 |
| bing | tablet | 1 | 5 | 1 | 831 | Iran | 0 |
| social | tablet | 0 | 3 | 12 | 324 | China | 0 |
| yahoo | tablet | 0 | 3 | 5 | 140 | Latvia | 0 |
| yahoo | tablet | 1 | 3 | 1 | 332 | France | 0 |

## 3.8 BETWEEN

The BETWEEN operator selects values within a given range and is inclusive: begin and end values are included. The values can be numbers, text, or dates. In the following example, we filter rows

from the `ecom` table where the visit count is between 1 and 3, and the `device` used to visit the website is `mobile`.

```
SELECT *
FROM ecom
WHERE n_visit BETWEEN 1 AND 3 AND device = 'mobile'
```

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| bing | mobile | 0 | 3 | 19 | 209 | Netherlands | 0 |
| direct | mobile | 1 | 2 | 1 | 501 | El Salvador | 0 |
| bing | mobile | 0 | 1 | 1 | 25 | Ireland | 0 |
| yahoo | mobile | 1 | 3 | 1 | 700 | Canada | 0 |
| direct | mobile | 0 | 1 | 13 | 234 | Indonesia | 0 |
| social | mobile | 0 | 1 | 2 | 44 | France | 0 |
| bing | mobile | 0 | 3 | 18 | 288 | Portugal | 1 |
| yahoo | mobile | 1 | 1 | 1 | 73 | Indonesia | 0 |
| google | mobile | 0 | 2 | 3 | 69 | Bahamas | 0 |
| direct | mobile | 1 | 1 | 1 | 423 | Mexico | 0 |

## 3.9 IN

The IN operator allows us to specify multiple values in a WHERE clause. It is a shorthand for multiple OR conditions. In the below example, we filter rows from the `ecom` table where the visit count is either 2 or 4 or 6 or 8 or 10. Instead of using multiple OR conditions, we use the IN operator.

```
SELECT *
FROM ecom
WHERE n_visit IN (2, 4, 6, 8, 10)
```

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| google | laptop | 1 | 10 | 1 | 693 | Czech Republic | 0 |
| yahoo | mobile | 1 | 10 | 1 | 75 | Bangladesh | 0 |
| direct | mobile | 1 | 10 | 1 | 908 | Indonesia | 0 |
| google | mobile | 1 | 6 | 1 | 208 | Czech Republic | 0 |
| direct | tablet | 0 | 6 | 12 | 132 | Estonia | 0 |

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| social | tablet | 0 | 2 | 1 | 28 | Namibia | 0 |
| direct | laptop | 1 | 2 | 1 | 384 | China | 0 |
| social | laptop | 1 | 4 | 1 | 850 | Ukraine | 0 |
| direct | tablet | 0 | 2 | 19 | 342 | Japan | 1 |
| google | mobile | 1 | 10 | 1 | 338 | Russia | 0 |

## 3.10 IS NULL

A field with a NULL value is a field with no value. If a field in a table is optional, it is possible to insert a new record or update a record without adding a value to this field. Then, the field will be saved with a NULL value. In the next example, we filter all rows from the `ecom` table where the `device` column has NULL values.

```
SELECT *
FROM ecom
WHERE device IS NULL
```

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|

## 3.11 LIKE

The LIKE operator is used to search for a specific pattern in a column. There are two wildcards used in conjunction with the LIKE operator:

- `%` : represents zero, one, or multiple characters
- `_` : represents a single character

In the following example, we filter all rows from the `ecom` table where the name of the `country` starts with `P`. We use `%` after `P` to indicate that it can be followed by any number or type of characters.

```
SELECT *
FROM ecom
WHERE country LIKE 'P%'
```

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| yahoo | mobile | 1 | 9 | 1 | 955 | Poland | 0 |
| yahoo | tablet | 0 | 5 | 8 | 80 | Philippines | 0 |
| bing | tablet | 0 | 5 | 16 | 368 | Peru | 1 |
| bing | laptop | 1 | 0 | 1 | 456 | Portugal | 0 |

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| yahoo | tablet | 1 | 9 | 1 | 706 | Philippines | 0 |
| direct | tablet | 1 | 8 | 1 | 814 | Philippines | 0 |
| social | laptop | 0 | 5 | 16 | 176 | Panama | 0 |
| yahoo | laptop | 1 | 4 | 1 | 928 | Peru | 0 |
| yahoo | tablet | 0 | 3 | 11 | 176 | Philippines | 0 |
| google | laptop | 1 | 9 | 1 | 869 | Poland | 0 |

Let us look at another example where we filter all rows from the `ecom` table where the name of the country should follow the below conditions:

- name can start with any character
- the second character must be `o`
- it can have any type or number of characters after the second character

```
SELECT *
FROM ecom
WHERE country LIKE '_o%'
```

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| yahoo | mobile | 1 | 9 | 1 | 955 | Poland | 0 |
| yahoo | laptop | 0 | 5 | 5 | 135 | South Africa | 0 |
| social | tablet | 0 | 7 | 10 | 290 | Colombia | 1 |
| yahoo | mobile | 0 | 8 | 9 | 225 | Colombia | 0 |
| direct | laptop | 1 | 8 | 1 | 658 | Colombia | 0 |
| bing | laptop | 1 | 0 | 1 | 456 | Portugal | 0 |
| yahoo | tablet | 0 | 7 | 11 | 264 | Somalia | 0 |
| google | tablet | 0 | 1 | 19 | 304 | Colombia | 0 |
| google | laptop | 1 | 9 | 1 | 869 | Poland | 0 |
| social | tablet | 1 | 1 | 1 | 566 | Portugal | 0 |

```
dbDisconnect(con)
```

## 3.12 Exercises

1. List the `referrer` and `device` of 5 visits with more than 8 page views (`n_visit > 8`), ordered by longest `duration` first.
2. How many rows are from `tablet` devices in countries starting with `P`?

3. Which distinct `device` values appear among bouncers (`bouncers = 1`)?

Worked solutions in `solutions/sql-basics.md` (attempt first).

# 4 SQL Advanced

## 4.1 Introduction

In this chapter, we will learn to

- aggregate data
- order data
- and group data

We will use the following R packages:

- DBI
- RSQLite
- dbplyr

All the data sets used in this chapter can be found here and code can be downloaded from here or run from `code/ch4-sqladvanced.R` in this repo.

## 4.2 Set Up

In order to be able to use the `ecom` data set for our case study, we will read it using the **readr** package and then copy it to an in memory database using the `copy_to()` function from **dplyr**.

```
# Canonical ecom: 1000 rows x 8 cols (referrer, device, bouncers, n_visit,
# n_pages, duration, country, purchase), from the local data/web.csv snapshot.
web_path <- if (file.exists("data/web.csv")) "data/web.csv" else "https://raw.git
hubusercontent.com/rsquaredacademy/datasets/master/web.csv"
ecom <- readr::read_csv(web_path, show_col_types = FALSE)
ecom <- dplyr::select(ecom, referrer, device, bouncers, n_visit, n_pages, duration,
country, purchase)
con <- DBI::dbConnect(RSQLite::SQLite(), ":memory:")
dplyr::copy_to(con, ecom)
```

## 4.3 Aggregate

Let us combine the aggregate statements with WHERE statement to filter data.

- SUM() : returns the total sum of a numeric column

```
SELECT SUM(n_visit) FROM ecom
```

| SUM(n_visit) |
|---|
| 4972 |

```sql
SELECT SUM(n_visit)
FROM ecom
WHERE n_visit > 5
```

| SUM(n_visit) |
|---|
| 3574 |

- AVG() : returns the average value of a numeric column

```sql
SELECT AVG(n_visit) FROM ecom
```

| AVG(n_visit) |
|---|
| 4.972 |

```sql
SELECT AVG(n_visit)
FROM ecom
WHERE country LIKE 'P%'
```

| AVG(n_visit) |
|---|
| 5.079137 |

- MAX() : returns the largest value of the selected column

```sql
SELECT MAX(n_visit) FROM ecom
```

| MAX(n_visit) |
|---|
| 10 |

```sql
SELECT MAX(n_visit)
FROM ecom
WHERE device = 'tablet'
```

| MAX(n_visit) |
|---|
| 10 |

- MIN() : returns the smallest value of the selected column

```sql
SELECT MIN(n_visit) FROM ecom
```

| MIN(n_visit) |
|---|
| 0 |

```sql
SELECT MIN(n_visit)
FROM ecom
WHERE duration BETWEEN 600 AND 900
```

| MIN(n_visit) |
|---|
| 0 |

## 4.4 Alias

SQL aliases are used to give a table, or a column in a table, a temporary name. They are often used to make column names more readable. An alias only exists for the duration of the query. Below are a few examples:

```
SELECT AVG(n_visit) AS avg_mobile
FROM ecom
WHERE device = 'mobile'
```

| avg_mobile |
|---|
| 5.479651 |

```
SELECT MAX(n_visit) AS max_visit
FROM ecom
```

| max_visit |
|---|
| 10 |

```
SELECT MIN(duration) AS min_duration
FROM ecom
```

| min_duration |
|---|
| 10 |

## 4.5 Order By

The ORDER BY keyword is used to sort the records in ascending or descending order. By default, the records are sorted in ascending order. Use the DESC keyword if you want to sort the records in descending order,

```
SELECT *
FROM ecom
ORDER BY country
```

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| social | laptop | 0 | 8 | 2 | 60 | Afghanistan | 0 |
| yahoo | laptop | 0 | 10 | 18 | 180 | Afghanistan | 0 |
| social | laptop | 1 | 2 | 1 | 274 | Afghanistan | 0 |
| direct | tablet | 1 | 10 | 1 | 682 | Afghanistan | 0 |

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| direct | tablet | 0 | 5 | 18 | 414 | Afghanistan | 1 |
| bing | tablet | 0 | 7 | 2 | 34 | Afghanistan | 0 |
| direct | laptop | 0 | 6 | 6 | 96 | Albania | 0 |
| google | laptop | 1 | 5 | 1 | 304 | Albania | 0 |
| bing | mobile | 0 | 2 | 2 | 54 | Albania | 0 |
| yahoo | laptop | 0 | 3 | 1 | 23 | Albania | 0 |

```
SELECT *
FROM ecom
ORDER BY duration
```

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| yahoo | tablet | 1 | 5 | 1 | 10 | Poland | 0 |
| social | laptop | 1 | 1 | 1 | 10 | Finland | 0 |
| yahoo | laptop | 0 | 0 | 1 | 12 | Indonesia | 0 |
| social | mobile | 1 | 2 | 1 | 12 | Botswana | 0 |
| direct | laptop | 0 | 0 | 1 | 13 | Czech Republic | 0 |
| direct | mobile | 0 | 10 | 1 | 17 | China | 0 |
| yahoo | tablet | 0 | 4 | 1 | 17 | Poland | 0 |
| yahoo | mobile | 0 | 7 | 1 | 19 | France | 0 |
| bing | laptop | 0 | 5 | 1 | 20 | Thailand | 0 |
| yahoo | mobile | 0 | 6 | 1 | 21 | Russia | 0 |

```
SELECT *
FROM ecom
ORDER BY n_visit DESC
```

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| google | laptop | 1 | 10 | 1 | 693 | Czech Republic | 0 |

| referrer | device | bouncers | n_visit | n_pages | duration | country | purchase |
|---|---|---|---|---|---|---|---|
| yahoo | mobile | 1 | 10 | 1 | 75 | Bangladesh | 0 |
| direct | mobile | 1 | 10 | 1 | 908 | Indonesia | 0 |
| google | mobile | 1 | 10 | 1 | 338 | Russia | 0 |
| google | mobile | 1 | 10 | 1 | 479 | Cameroon | 0 |
| direct | mobile | 0 | 10 | 1 | 17 | China | 0 |
| direct | tablet | 0 | 10 | 3 | 57 | China | 0 |
| social | mobile | 0 | 10 | 11 | 242 | Argentina | 1 |
| direct | mobile | 1 | 10 | 1 | 675 | Indonesia | 0 |
| google | tablet | 1 | 10 | 1 | 86 | China | 0 |

## 4.6 Group By

The GROUP BY statement is used with aggregate functions (COUNT, MAX, MIN, SUM, AVG) to group the result by one or more columns.

```
SELECT device,
count(*) AS visits
FROM ecom
GROUP BY device
ORDER by visits DESC
```

| device | visits |
|---|---|
| mobile | 344 |
| tablet | 331 |
| laptop | 325 |

```
SELECT device, MAX(duration) AS max_duration
FROM ecom
GROUP BY device
ORDER by max_duration DESC
```

| device | max_duration |
|---|---|
| tablet | 999 |
| laptop | 997 |
| mobile | 994 |

```
dbDisconnect(con)
```

### 4.7 Exercises

1. Compute the average pages per visit (`AVG(n_pages)`) for each `referrer`, ordered highest average first. Which referrer tops the list?
2. What is the longest `duration` among visits that made a purchase (`purchase = 1`)?
3. Among `mobile` visits only, count visits by `country` and show the top 5. Which country is first?

Worked solutions in `solutions/sql-advanced.md` (attempt first).

## 5 JOINs

### 5.1 Introduction

In this chapter, we will learn to:

- combine rows from two tables with `INNER JOIN`
- keep unmatched rows with `LEFT JOIN`
- avoid the SQLite `RIGHT` / `FULL OUTER JOIN` trap
- do the same joins from R with dbplyr `inner_join()` / `left_join()`
- filter with `semi_join()` and `anti_join()`

We will use the following R packages:

- DBI
- RSQLite
- dbplyr

All the data sets used in this chapter can be found here and code can be run from `code/ch5-joins.R` in this repo.

### 5.2 Set Up

So far we have worked with a single table. Real databases keep related data in separate tables. Our `data/ecom.sqlite` file holds two small related tables besides `ecom`:

- `customers`: 20 rows, one per customer (`customer_id` primary key, `country`)
- `orders`: 20 rows, one per order (`order_id` primary key, `customer_id` foreign key into `customers`, `amount`)

Six customers (4, 16–20) have placed no orders, and one order points at a `customer_id` (99) that does not exist. These deliberate mismatches make the difference between join types visible.

Let us connect directly to the SQLite file:

```r
con <- DBI::dbConnect(RSQLite::SQLite(), "data/ecom.sqlite")
```

A quick look at both tables:

```sql
SELECT * FROM customers LIMIT 5
```

| customer_id | country |
|---|---|
| 1 | United States |
| 2 | United Kingdom |
| 3 | Germany |
| 4 | France |
| 5 | Canada |

```sql
SELECT * FROM orders LIMIT 5
```

| order_id | customer_id | amount |
|---|---|---|
| 101 | 1 | 459.11 |
| 102 | 1 | 469.80 |
| 103 | 2 | 157.35 |
| 104 | 3 | 418.61 |
| 105 | 3 | 328.04 |

## 5.3 INNER JOIN

`INNER JOIN` returns only the rows that match in **both** tables. Customers without orders, and orders without customers, are dropped.

```sql
SELECT c.customer_id, c.country, o.order_id, o.amount
FROM customers c
INNER JOIN orders o ON c.customer_id = o.customer_id
ORDER BY c.customer_id, o.order_id
LIMIT 6
```

| customer_id | country | order_id | amount |
|---|---|---|---|
| 1 | United States | 101 | 459.11 |
| 1 | United States | 102 | 469.80 |
| 2 | United Kingdom | 103 | 157.35 |
| 2 | United Kingdom | 118 | 76.39 |
| 3 | Germany | 104 | 418.61 |
| 3 | Germany | 105 | 328.04 |

How many of the 20 orders survive? Nineteen – order 120 belongs to the non-existent customer 99 and is excluded:

```
SELECT COUNT(*) AS n FROM customers c
INNER JOIN orders o ON c.customer_id = o.customer_id
```

| n |
|---|
| 19 |

## 5.4 LEFT JOIN

`LEFT JOIN` keeps **every** row from the left table and fills `NULL` where the right table has no match. This is how you find customers who never ordered:

```
SELECT c.customer_id, c.country, o.order_id, o.amount
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id
WHERE o.order_id IS NULL
ORDER BY c.customer_id
```

| customer_id | country | order_id | amount |
|---|---|---|---|
| 4 | France | NA | NA |
| 16 | Italy | NA | NA |
| 17 | Portugal | NA | NA |
| 18 | Greece | NA | NA |
| 19 | Ireland | NA | NA |
| 20 | Belgium | NA | NA |

The full left join has 25 rows: 19 matched order rows plus one `NULL` row for each of the 6 customers without orders.

## 5.5 The SQLite RIGHT / FULL Trap

Some tutorials teach `RIGHT JOIN` or `FULL OUTER JOIN`. Do not use them with RSQLite: the SQLite versions bundled with RSQLite (including recent ones) reject both with a syntax error such as `near "RIGHT": syntax error`. Postgres and other production databases accept them, so queries written against SQLite will fail silently only when you move them.

The portable workaround is to swap the tables and use `LEFT JOIN`:

- instead of `A RIGHT JOIN B`, write `B LEFT JOIN A`
- instead of `A FULL OUTER JOIN B`, union a `LEFT JOIN` in each direction

When in doubt, `LEFT JOIN` is the only outer join you need for this book.

## 5.6 JOINs with dbplyr

The same joins work from R without writing SQL. Reference both tables and use the dplyr join verbs:

```
customers <- dplyr::tbl(con, "customers")
orders <- dplyr::tbl(con, "orders")

inner_join(customers, orders, by = "customer_id")
```

```
# A query:  ?? x 4
# Database: sqlite 3.53.3 [/home/runner/work/rdbsql/rdbsql/data/ecom.sqlite]
   customer_id country        order_id amount
         <int> <chr>             <int>  <dbl>
 1           1 United States       101  459.
 2           1 United States       102  470.
 3           2 United Kingdom      103  157.
 4           2 United Kingdom      118   76.4
 5           3 Germany             104  419.
 6           3 Germany             105  328.
 7           3 Germany             106  269.
 8           5 Canada              107  374.
 9           5 Canada              119  248
10           6 Australia           108   84.6
11           7 Brazil              109  335.
12           8 Japan               110  358.
13           9 Mexico              111  240.
14          10 Netherlands         112  365.
15          11 Sweden              113  469.
16          12 Norway              114  143.
17          13 Denmark             115  242.
18          14 Finland             116  471.
19          15 Poland              117  490.
```

Use `show_query()` to see the SQL dbplyr generates – it is the `INNER JOIN` you just wrote by hand:

```
q_inner <- inner_join(customers, orders, by = "customer_id")
dplyr::show_query(q_inner)
## <SQL>
## SELECT `customers`.*, `order_id`, `amount`
## FROM `customers`
## INNER JOIN `orders`
##   ON (`customers`.`customer_id` = `orders`.`customer_id`)
```

The left join works the same way:

```
q_left <- left_join(customers, orders, by = "customer_id")
dplyr::show_query(q_left)
```

```
## <SQL>
## SELECT `customers`.*, `order_id`, `amount`
## FROM `customers`
## LEFT JOIN `orders`
##   ON (`customers`.`customer_id` = `orders`.`customer_id`)
```

## 5.7 semi_join and anti_join

Two filtering joins have no direct SQL keyword but are everyday tools:

- `semi_join()` keeps rows of `customers` that **have** a match in `orders` (14 customers). The generated SQL uses `WHERE EXISTS`.
- `anti_join()` keeps rows of `customers` that **lack** a match in `orders` (the same 6 customers from the `LEFT JOIN` check above). The generated SQL uses `WHERE NOT EXISTS`.

```
q_semi <- semi_join(customers, orders, by = "customer_id")
dplyr::show_query(q_semi)
## <SQL>
## SELECT `customers`.*
## FROM `customers`
## WHERE EXISTS (
##   SELECT 1 FROM `orders`
##   WHERE (`customers`.`customer_id` = `orders`.`customer_id`)
## )
dplyr::collect(q_semi)
## # A tibble: 14 x 2
##    customer_id country
##          <int> <chr>
##  1           1 United States
##  2           2 United Kingdom
##  3           3 Germany
##  4           5 Canada
##  5           6 Australia
##  6           7 Brazil
##  7           8 Japan
##  8           9 Mexico
##  9          10 Netherlands
## 10          11 Sweden
## 11          12 Norway
## 12          13 Denmark
## 13          14 Finland
## 14          15 Poland
```

```
q_anti <- anti_join(customers, orders, by = "customer_id")
dplyr::show_query(q_anti)
## <SQL>
## SELECT `customers`.*
```

```
## FROM `customers`
## WHERE NOT EXISTS (
##   SELECT 1 FROM `orders`
##   WHERE (`customers`.`customer_id` = `orders`.`customer_id`)
## )
dplyr::collect(q_anti)
## # A tibble: 6 x 2
##   customer_id country
##         <int> <chr>
## 1           4 France
## 2          16 Italy
## 3          17 Portugal
## 4          18 Greece
## 5          19 Ireland
## 6          20 Belgium
```

Rule of thumb: use `semi_join()` / `anti_join()` when you want to filter one table by membership in another without adding the other table's columns.

```
dbDisconnect(con)
```

### 5.8 Exercises

1. List the `customer_id` and `country` of customers who placed no orders. Write it both as a `LEFT JOIN ... IS NULL` query and as an `anti_join()`.
2. Compute total order `amount` per customer (highest first) with an `INNER JOIN` + `GROUP BY`. Which customer spent the most?
3. Find the orphan order: which `order_id` points at a non-existent customer?

Worked solutions in `solutions/joins.md` (attempt first).

## 6 DBI Cheat Sheet

One-page reference: connect, query, disconnect. Printable PDF companion to this book (CC BY-NC-SA 4.0).

### 6.1 Connect, query, disconnect

```
library(DBI)
con <- dbConnect(RSQLite::SQLite(), ":memory:")  # or duckdb::duckdb()
dbListTables(con)                                # what is in the database?
dbReadTable(con, "ecom")                         # whole table (small only!)
dbGetQuery(con, "SELECT * FROM ecom LIMIT 10")   # SQL, first rows
res <- dbSendQuery(con, "SELECT * FROM ecom")    # big tables: fetch in batches
x <- dbFetch(res, n = 100)
dbClearResult(res)                               # always clear when done
dbDisconnect(con)                                # always disconnect
```

### 6.2 Write data

```
dbWriteTable(con, "trial", df)                  # create (overwrite / append args)
dbExecute(con, "INSERT INTO trial (x) VALUES (1)") # statements, no cleanup needed
dbRemoveTable(con, "trial")                       # delete table
```

### 6.3 Inspect a query

`dbHasCompleted()` `dbGetStatement()` `dbGetRowCount()` `dbGetRowsAffected()` `dbColumnInfo()` — status, SQL, counts, column types of a `dbSendQuery()` result.

### 6.4 dbplyr in 4 verbs

```
e <- dplyr::tbl(con, "ecom")        # lazy reference, pulls nothing
e |> dplyr::filter(duration > 300)  # WHERE
e |> dplyr::group_by(device) |>     # GROUP BY + aggregates
  dplyr::summarise(avg = mean(duration))
dplyr::show_query(q)                # see the generated SQL
dplyr::collect(q)                   # pull results into a tibble
inner_join(a, b) / left_join(a, b)  # JOINs; semi/anti_join filter
```

# Appendices

# 7 Production R Patterns

So far every chapter used an in-memory SQLite database. That was deliberate: zero setup, everything reproducible. Real jobs look different — a Postgres server with credentials, concurrent writers, and tables far bigger than memory.

The good news: almost everything transfers. `DBI` is a common interface, so `dbGetQuery()`, `dbWriteTable()`, and `dbDisconnect()` work the same everywhere. This appendix covers the six things that change when you leave SQLite.

We will use the following R packages:

- DBI
- RSQLite

Sections that need a live server are display-only (`eval=FALSE`) — read them, do not run them. Sections that run on SQLite execute normally.

### 7.1 Connection Strings

Only the `dbConnect()` call changes per backend. Everything after it is identical DBI code:

```
# Postgres (production default for most R teams)
con <- DBI::dbConnect(RPostgres::RPostgres(),
  host = "db.internal", dbname = "analytics",
  user = Sys.getenv("DB_USER"), password = Sys.getenv("DB_PASS"))

# MariaDB / MySQL
```

```
con <- DBI::dbConnect(RMariaDB::MariaDB(),
  host = "db.internal", dbname = "analytics",
  user = Sys.getenv("DB_USER"), password = Sys.getenv("DB_PASS"))

# Anything with an ODBC driver (SQL Server, Oracle, Snowflake)
con <- DBI::dbConnect(odbc::odbc(), dsn = "warehouse",
  uid = Sys.getenv("DB_USER"), pwd = Sys.getenv("DB_PASS"))
```

For the full backend matrix (BigQuery, Snowflake, Oracle DSNs, enterprise security), see Posit Databases using R. This book stays backend-agnostic on purpose: learn the patterns here once, then look up one connection string.

## 7.2 Credentials: env vars + keyring

Never paste passwords into scripts. Two rules, in order of preference:

```
# 1. Environment variables (works everywhere: laptop, CI, Posit Connect)
Sys.setenv(DB_USER = "analyst")  # once per session, never committed
con <- DBI::dbConnect(RPostgres::RPostgres(), password = Sys.getenv("DB_PASS"))

# 2. OS credential store for interactive work (macOS Keychain, Windows
#    Credential Manager, Linux Secret Service)
keyring::key_set("warehouse", "analyst")  # prompts once, stores securely
pwd <- keyring::key_get("warehouse", "analyst")
```

If a password appears in a committed file, rotate it — editing history does not un-leak it.

## 7.3 Parameterized Queries

Build SQL with placeholders, not `paste0()`. String-pasting user input into queries is how SQL injection happens (especially in Shiny apps). This section executes — placeholders work on SQLite too:

```
device <- "mobile"
q <- DBI::sqlInterpolate(con,
  "SELECT COUNT(*) AS n FROM ecom WHERE device = ?dev",
  dev = device)
q
```

```
<SQL> SELECT COUNT(*) AS n FROM ecom WHERE device = 'mobile'
```

```
dbGetQuery(con, q)
```

```
    n
1 344
```

`sqlInterpolate()` quotes and escapes `?dev` for the active backend, so the same code is safe on Postgres. For composing queries programmatically, the equivalent is `glue::glue_sql("... {dev}", .con = con)` — note `.con`, which tells `glue` which dialect to quote for.

## 7.4 Transactions

When several writes must succeed or fail together (debit one account, credit another), wrap them in a transaction. `dbWithTransaction()` commits on success and rolls back automatically if any step errors:

```
dbWriteTable(con, "trial", data.frame(id = 1:3, x = c("a", "b", "c")))
dbWithTransaction(con, {
  dbExecute(con, "INSERT INTO trial (id, x) VALUES (4, 'd')")
  dbExecute(con, "DELETE FROM trial WHERE id = 1")
})
```

```
[1] 1
```

```
dbGetQuery(con, "SELECT * FROM trial ORDER BY id")
```

```
  id x
1  2 b
2  3 c
3  4 d
```

A failed transaction leaves the table untouched — verify by forcing an error inside the block and re-reading the table. Prefer `dbWithTransaction()` over manual `dbBegin()` / `dbCommit()` / `dbRollback()`: the manual trio leaks open transactions when an error interrupts your script.

## 7.5 Writing Data at Scale

`copy_to()` (used throughout this book) is a demo convenience: it uploads an in-memory data frame with guessed column types. For real loads:

```
# Declare types explicitly instead of letting the database guess
DBI::dbCreateTable(con, "events",
  fields = c(id = "INTEGER", ts = "TIMESTAMP", label = "TEXT"))

# Control large uploads: overwrite once, append after
DBI::dbWriteTable(con, "events", january, overwrite = TRUE)
DBI::dbWriteTable(con, "events", february, append = TRUE)

# Speed repeated lookups with an index (or a view for saved queries)
DBI::dbExecute(con, "CREATE INDEX idx_events_ts ON events (ts)")
```

Rule of thumb: `copy_to()` for classroom demos, `dbWriteTable()` with `overwrite` / `append` for pipelines, `dbCreateTable()` + explicit types when the table must survive longer than your session.

## 7.6 Connections in Shiny: pool

A Shiny app that calls `dbConnect()` per session exhausts the database under load and leaks idle connections. The pool package checks connections out per reactive flush and returns them after:

```
pool <- pool::dbPool(RPostgres::RPostgres(), dbname = "analytics",
  host = "db.internal", user = Sys.getenv("DB_USER"))
```

```
# use pool exactly like con, then on app stop:
pool::poolClose(pool)
```

If you take one thing from this appendix into production Shiny work, take `pool`.

```
dbDisconnect(con)
```

Further reading: `vignette("DBI-advanced")` (transactions, quoting, bulk operations) and the DBI specification for backend authors. Runnable code for this appendix lives in `code/ch6-production.R` (server sections included as comments — uncomment with your credentials).

# 8 DuckDB + Parquet

SQLite was the right default for learning: zero setup, same DBI code everywhere. DuckDB is its analytical twin — an in-process engine (no server, like SQLite) built for fast aggregate queries over large files. The pitch for R users: the DBI code you already know runs unchanged, and you can query Parquet or CSV files directly without loading them into memory.

> All chunks in this appendix are display-only (`eval=FALSE`): they need `install.packages("duckdb")` once, which the book's automated builds skip. Run them locally — every line works as written. Runnable code lives in `code/ch7-duckdb.R` (skips gracefully when `duckdb` is not installed).

## 8.1 Connect: one Line Changes

```
# install.packages("duckdb")  # once; high-speed analytical engine
con_duck <- DBI::dbConnect(duckdb::duckdb())
DBI::dbWriteTable(con_duck, "ecom", ecom)
DBI::dbGetQuery(con_duck, "SELECT referrer, AVG(duration) FROM ecom GROUP BY
referrer")
```

Compare with Chapter 1: `RSQLite::SQLite()` became `duckdb::duckdb()`, and nothing else moved. `dbListTables()`, `dbGetQuery()`, `dbWriteTable()`, `dbDisconnect()` — the whole DBI vocabulary carries over, which is exactly why the book taught SQLite first.

## 8.2 Query Files Without Loading Them

DuckDB's signature trick: `FROM 'file'` reads Parquet or CSV straight off disk, pushing filters and aggregates down so R never holds the full data:

```
# Aggregate a 2 GB Parquet file from ~50 MB of R memory
DBI::dbGetQuery(con_duck, "
  SELECT country, SUM(amount) AS revenue
  FROM 'orders.parquet'
  WHERE year = 2025
  GROUP BY country
  ORDER BY revenue DESC
  LIMIT 10")
```

```
# Same for CSVs — handy for the raw logs before they become Parquet
DBI::dbGetQuery(con_duck, "
  SELECT COUNT(*) AS n FROM read_csv('logs/*.csv', header = true)")
```

Write results back out the same way (`COPY ... TO 'out.parquet'`), and you have a full extract-transform pipeline with no database server and no RAM anxiety. Pair with Arrow (`arrow::read_parquet()`, `arrow::write_parquet()`) when you want the files in R as well as SQL.

## 8.3 dbplyr on DuckDB

Lazy evaluation works exactly as in Chapter 2 — `tbl()`, verbs, `show_query()`, `collect()` — with DuckDB as the execution engine:

```
e <- dplyr::tbl(con_duck, "ecom")
q <- e |>
  dplyr::filter(duration > 300) |>
  dplyr::group_by(device) |>
  dplyr::summarise(avg = mean(duration))
dplyr::show_query(q)  # DuckDB SQL this time, same dplyr code
dplyr::collect(q)
```

Two habits carry over from Chapter 2: use `show_query()` to check what dbplyr generates (window functions and custom SQL still deserve hand-written queries via `dplyr::sql()`), and prefer `dplyr::compute()` over `collect()` for intermediate results too big for memory — `compute()` materializes them as a DuckDB table instead of an R tibble.

```
DBI::dbDisconnect(con_duck)
```

## 8.4 SQLite or DuckDB?

| | SQLite (Chapters 1–5) | DuckDB (this appendix) |
|---|---|---|
| Shape | Row-oriented, transactional | Column-oriented, analytical |
| Best at | Lookups, inserts, app state | Aggregates, scans, Parquet |
| Files | `.sqlite` | `.parquet`, `.csv`, in-memory |
| Scale | Fits comfortably | Bigger than RAM, via pushdown |
| R path | `RSQLite::SQLite()` | `duckdb::duckdb()` |

Default to SQLite for application data and DuckDB for analysis. When your analysis outgrows DuckDB on one machine, the next step up is R4DS Ch. 21 — the funnel this book was built to feed.